Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

1. Administrative & Study Identification

Full Study Title:	Phase Ib/II Trial of Pembrolizumab (MK-3475) Combination Therapies in Metastatic Castration-Resistant Prostate Cancer (mCRPC) (KEYNOTE-365)
Short Title:	Study of Pembrolizumab (MK-3475) Combination Therapies in Metastatic Castration-Resistant Prostate Cancer (MK-3475-365/KEYNOTE-365)
Protocol Number:	19839414555992495
NCT Number:	NCT02861573
Sponsor Name:	Merck
Investigational Product:	etoposide
Phase:	PHASE1, PHASE2
Medical Monitor:	[Not Provided in Posting]

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective:	1. Percentage of Participants With a Decrease of ?50% in Prostatic Specific Antigen (PSA) 2. Number of Participants with Adverse Events (AEs) 3. Number of Participants Discontinuing Study Drug Due to AEs
Secondary Obj.:	1. Disease Control Rate (DCR) Based on Prostate Cancer Working Group 3 (PCWG3)-modified RECIST 1.1 Assessed by BICR 2. Overall Survival (OS) 3. Duration of Response (DOR) Based on PCWG3-modified RECIST 1.1 Assessed by BICR

2.2 Background & Rationale

The purpose of this study is to assess the safety and efficacy of pembrolizumab (MK-3475) combination therapy in participants with metastatic castration resistant prostate cancer (mCRPC). There will be ten cohorts in this study: Cohort A will receive pembrolizumab + olaparib, Cohort B will receive pembrolizumab + docetaxel + prednisone, Cohort C will receive pembrolizumab + enzalutamide, Cohort D will receive pembrolizumab + abiraterone + prednisone Cohort E will receive pembrolizumab+lenvatinib, Cohort F will receive pembrolizumab+lenvatinib, Cohort G will receive pembrolizumab/vibostolimab coformulation (MK-7684A), Cohort H will receive pembrolizumab/vibostolimab coformulation, Cohort I will receive pembrolizumab+carboplatin+etoposide in Arm 1 and carboplatin+etoposide in Arm 2 and Cohort J will receive belzutifan in Arm1 and Pembrolizumab+belzutifan in Arm 2. Outcome measures will be assessed individually for each cohort.

3. Study Design & Methodology

Design Framework:	Type: INTERVENTIONAL, Phase: PHASE1, PHASE2, Status: RECRUITING
Target N:	1200

4. Subject Selection (Eligibility Criteria)

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4.1 Inclusion Criteria

* For Cohorts A, B, C, D, E, G, J: Has histologically- or cytologically-confirmed adenocarcinoma of the prostate without small cell histology

* For Cohorts F, H, I: Has t-NE or de novo metastatic prostate cancer defined by ?1% neuroendocrine cells that are located in discrete regions of a recent biopsy specimen from a metastasis as determined by the investigational site and confirmed by central histology review prior to enrollment. Epstein criteria of neuroendocrine differentiation in prostate cancer is used for eligibility. Specimens must have one of the morphologies of Small cell carcinoma or Large cell neuroendocrine carcinoma (LCNEC) or Mixed (small or large cell) NE carcinoma - acinar adenocarcinoma with positive IHC confirmed by central pathology review
* Is able to provide tumor tissue from a site not previously irradiated as follows: Cohorts A, E, G and J: must provide a core or excisional biopsy from soft tissue or bone biopsy within 1 year of screening and after developing mCRPC; Cohort B: must provide an archival tumor tissue sample or tumor tissue from a newly obtained core or excisional biopsy from soft tissue if the lesion is clinically accessible; Cohorts C and D with soft tissue disease: must provide a core or excisional biopsy from a soft tissue lesion if clinically accessible within 1 year of screening and after developing mCRPC and an archival specimen if available; and Cohorts F, H, and I must provide a core or excisional biopsy from soft tissue or a bone biopsy. For de novo metastatic neuroendocrine prostate participants, biopsies must be performed within 1 year of screening. Participants with bone metastasis only must provide an archival tumor tissue specimen
* Has prostate cancer progression within 6 months prior to screening, as determined by the investigator, by means of one of the following: PSA progression as defined by a minimum of 2 rising PSA levels with an interval of ?1 week between each assessment where the PSA value at screening should be ?2 ng/mL; radiographic disease progression in soft tissue based on Response Evaluation Criteria In Solid Tumors Version 1.1 criteria with or without PSA progression; radiographic disease progression in bone defined as the appearance of 2 or more new bone lesions on bone scan with or without PSA progression. Participants with de novo neuroendocrine prostate cancer will not need to provide evidence of progression within 6 months
* Has ongoing androgen deprivation with serum testosterone \<50 ng/dL (\<2.0 nM). Treatment with luteinizing hormone-releasing hormone agonists or antagonists for all cohorts must have been initiated ?4 weeks prior to first dose of study therapy and must be continued throughout the study. Participants with de novo metastatic NE prostate cancer will not be required to have been previously treated with androgen deprivation therapy (ADT). ADT must be started in these participants by the time of treatment allocation/randomization
* Participants receiving bone resorptive therapy (including, but not limited to bisphosphonate or receptor activator of nuclear factor kappa-? ligand inhibitor) must be on stable doses for ?4 weeks prior to first dose of study therapy
* Must be abstinent from heterosexual intercourse, refrain from donating sperm, or agree to use contraception (unless confirmed to be azoospermic) during the intervention period starting with the first dose of study therapy. The length of time required to continue contraception after the last dose of study intervention for each study intervention is as follows: 7 days for abiraterone acetate and lenvatinib; 30 days for enzalutamide; and 95 days for olaparib, docetaxel, and carboplatin/etoposide. No contraception measures are required during and after the intervention period for pembrolizumab/vibostolimab coformulation
* Has a performance status of 0, 1, or 2 on the Eastern Cooperative Oncology Group (ECOG) Performance Scale for Cohorts A and C and a performance status of 0 or 1 for Cohorts B, D, E, F, G, H, I and J within 10 days of study start

* For Cohort A: Has received docetaxel for mCRPC. Prior treatment with 1 other chemotherapy for mCRPC is allowed. Up to 2 second-generation hormonal manipulations (e.g., abiraterone acetate and/or enzalutamide) are allowed. Prior docetaxel for metastatic hormone-sensitive prostate cancer is allowed if ?4 weeks have elapsed from the last dose of docetaxel prior to day 1 of Cycle 1
* For Cohort B: Has received prior treatment with either abiraterone acetate or enzalutamide (but not both) in the prechemotherapy mCRPC state. Participants in Cohort B must have received at least 4 weeks of either abiraterone or enzalutamide treatment (but not both) who failed treatment or became intolerant of the drug
* For Cohort C: Has received prior treatment with abiraterone acetate in the pre-chemotherapy mCRPC state without prior enzalutamide. Participants in Cohort C must have received at least 4 weeks of abiraterone treatment who failed treatment or become intolerant of the drug. Participants who received abiraterone acetate in the hormone-sensitive state will not be eligible
* For Cohort D: Has not received chemotherapy for mCRPC and has either not had prior second generation hormonal manipulation for mCRPC OR has previously been treated with enzalutamide for mCRPC and failed treatment or has become intolerant of the drug. Prior docetaxel for metastatic hormone-sensitive prostate cancer is allowed if ?4 weeks have elapsed from the last dose of docetaxel. Prior treatment with abiraterone acetate in the hormone-sensitive metastatic setting is allowed as long as there was no progression on this agent and abiraterone acetate was not discontinued due to adverse events (AEs)
* For Cohorts E, G and J: Has received docetaxel for mCRPC. Prior treatment with 1 other chemotherapy for mCRPC is allowed. Up to 2 second-generation hormonal manipulations (eg, abiraterone acetate, enzalutamide, apalutamide, darolutamide or other next-generation hormonal agents \[NHA\]) are allowed. Participants who received prior ketoconazole for metastatic disease may be enrolled. If docetaxel chemotherapy is used more than once (eg, once for metastatic hormone-sensitive and once for mCRPC), it will be considered as 1 therapy. Prior docetaxel for metastatic hormone-sensitive prostate cancer (mHSPC) is allowed if ?4 weeks have elapsed from the last dose of docetaxel prior to Day 1 of Cycle 1
* For Cohort F, H, and I: Participants must have received prior treatment with androgen deprivation therapy (ADT) for metastatic disease. Prior treatment with up to a total of 2 chemotherapies for mCRPC is allowed, as well as up to 2 second-generation hormonal manipulations for mCRPC. Participants who received prior ketoconazole for metastatic disease may be enrolled. Docetaxel for mHSPC is allowed in addition to docetaxel for mCRPC. If docetaxel chemotherapy is used more than once (eg, once for metastatic hormone-sensitive and once for mCRPC), it will be considered as 1 therapy

4.2 Exclusion Criteria

* Has had a prior anticancer monoclonal antibody (mAb) within 4 weeks prior to first dose study therapy or who has not recovered (i.e., Grade ?1 or at baseline) from AEs due to mAbs administered \>4 weeks earlier
* Has had prior chemotherapy, targeted small molecule therapy abiraterone treatment, enzalutamide treatment, or radiation therapy within 2 weeks prior to first dose of study therapy or who has not recovered (ie, Grade ?1 or at baseline) from AEs due to a previously administered agent
* Is currently participating in and receiving study therapy or has participated in a study of an investigational agent and received study drug or used an investigational device within 4 weeks of treatment allocation/randomization
* Has a diagnosis of immunodeficiency or is receiving systemic steroid therapy or any other form of immunosuppressive therapy within 7 days prior to treatment allocation/randomization

* Has had prior treatment with therapeutic radiopharmaceuticals for prostate cancer, such as Radium-223 or Leutitium-177, within 4 weeks prior to the first dose of trial treatment
* Has an active autoimmune disease that has required systemic treatment in past 2 years
* Has a history of (noninfectious) pneumonitis/interstitial lung disease that required steroids or current pneumonitis/interstitial lung disease
* Has previously participated in any other pembrolizumab trial, or received prior therapy with an anti-programmed cell death 1 (anti-PD-1), anti-programmed cell death ligand 1 (anti-PD-L1), and anti-programmed cell death ligand 1 (anti-PD-L2)
* Has a known history of Human Immunodeficiency Virus (HIV)
* Has known active Hepatitis B or Hepatitis C
* Has received a live vaccine or live-attenuated vaccine within 30 days of the first dose of study therapy
* Has known active central nervous system metastases and/or carcinomatous meningitis
* Has a "superscan" bone scan defined as an intense symmetric activity in the bones and diminished renal parenchymal activity on baseline bone scan such that the presence of additional metastases in the future could not be evaluated
* Has had prior solid, organ or bone marrow transplant
* For Cohort A: Has experienced a seizure or seizures within 6 months of study start or is currently being treated with cytochrome P450 enzyme (CYP) inducing anti-epileptic drugs for seizures
* For Cohort A: Is currently receiving strong or moderate inhibitors of CYP3A4 including azole antifungals; macrolide antibiotics; or protease inhibitors
* For Cohort A: Is currently receiving strong or moderate inducers of CYP3A4
* For Cohort A: Has myelodysplastic syndrome
* For Cohort A: Has symptomatic congestive heart failure (New York Heart Association Class III or IV heart disease), unstable angina pectoris, cardiac arrhythmia, or uncontrolled hypertension
* For Cohort B: Has received prior treatment with docetaxel or another chemotherapy agent for metastatic prostate cancer
* For Cohort B: Has peripheral neuropathy Common Terminology Criteria for Adverse Events ?2 except due to trauma
* For Cohort B: Has ascites and/or clinically significant pleural effusion
* For Cohort B:Has symptomatic congestive heart failure (New York Heart Association Class III or IV heart disease)
* For Cohort B: Is currently receiving any of the following classes of inhibitors of CYP3A4: azole antifungals; macrolide antibiotics; or protease inhibitors
* For Cohort C: Has received prior chemotherapy for mCRPC. Prior docetaxel for metastatic hormone-sensitive prostate cancer is allowed if ?4 weeks elapsed from last dose of docetaxel. Participants who received abiraterone acetate in the hormone-sensitive state will not be eligible
* For Cohort C: Has a history of seizure or any condition that may predispose to seizure (including, but not limited to prior cerebrovascular accident, transient ischemic attack, or brain arteriovenous malformation; or intracranial masses such as a schwannoma or meningioma that is causing edema or mass effect)
* For Cohort C:Has known or suspected brain metastasis or leptomeningeal carcinomatosis
* For Cohort C: Has a history of loss of consciousness within 12 months of the screening visit
* For Cohort C: Has hypotension (systolic blood pressure \<86 millimeters of mercury \[mmHg\]) or uncontrolled hypertension (systolic blood pressure \>170 mmHg or diastolic blood pressure \>105 mmHg) at the screening visit

* For Cohort C: Has received treatment with 5-? reductase inhibitors (e.g., finasteride, dutasteride), estrogens, and/or cyproterone within 4 weeks prior to Cycle 1
* For Cohort C: Has a history of prostate cancer progression on ketoconazole
* For Cohort D: Has received prior treatment with docetaxel or another chemotherapy agent for metastatic prostate cancer
* For Cohort D: Has progressed on prior abiraterone acetate for treatment of castration sensitive or resistant metastatic prostate cancer
* For Cohort D: Has discontinued prior treatment with abiraterone acetate due to AEs
* For Cohort D: Has previously been treated with ketoconazole for prostate cancer for \>7 days
* For Cohort D: Has received prior systemic treatment with an azole drug (eg, fluconazole, itraconazole) within 4 weeks of Cycle 1, Day 1
* For Cohort D: Has uncontrolled hypertension (systolic BP ? 160 mm Hg or diastolic BP ? 95 mm Hg)
* For Cohort D: Has a history of pituitary or adrenal dysfunction
* For Cohort D: Has clinically significant heart disease as evidenced by myocardial infarction, or arterial thrombotic events in the past 6 months, severe or unstable angina, or New York Heart Association Class II-IV heart disease or cardiac ejection fraction measurement of \<50% at baseline
* For Cohort D: Has atrial fibrillation, or other cardiac arrhythmia requiring therapy
* For Cohort D: Has a history of chronic liver disease
* For Cohort D: Is currently receiving strong CYP3A4 inducers (eg, phenytoin, carbamazepine, rifampin, rifabutin, rifapentine, phenobarbital) during abiraterone acetate treatment, CYP2D6 substrates with a narrow therapeutic index (for example thioridazine), or CYP2C8 substrates with a narrow therapeutic index (for example pioglitazone)
* For Cohorts E and F: Has a left ventricular ejection fraction (LVEF) below the institutional (or local laboratory) normal range, as determined by multigated acquisition (MUGA) or echocardiogram (ECHO)
* For Cohorts E and F: Has radiographic evidence of encasement or invasion of a major blood vessel, or of intratumoral cavitation
* For Cohorts E and F: Has prolongation of QT interval by Fredericia (QTcF) interval to \>480 milliseconds
* For Cohorts E and F: Has had major surgery within 3 weeks prior to first dose of study interventions
* For Cohorts E and F: Has pre-existing ?Grade 3 gastrointestinal or non-gastrointestinal fistula
* For Cohorts E and F: Has had significant cardiovascular impairment within 12 months of the first dose of study intervention, such as history of New York Heart Association \>Class II congestive heart failure, unstable angina, myocardial infarction, or cerebrovascular accident (CVA)/stroke, cardiac revascularization procedure, or cardiac arrhythmia associated with hemodynamic instability
* For Cohorts E and F: Has active hemoptysis (bright red blood of at least 0.5 teaspoon) within 3 weeks prior to the first dose of lenvatinib
* For Cohorts E and F: Has gastrointestinal malabsorption or any other condition that might affect the absorption of lenvatinib
* For Cohorts G, H, and I: Has had a severe hypersensitivity reaction to treatment with another monoclonal antibody
* For Cohorts G, H, and I: Has symptomatic ascites or pleural effusion
* For Cohort I: Has clinically active diverticulitis, intra-abdominal abscess, gastrointestinal obstruction, and/or abdominal carcinomatosis

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* For Cohort I: Has received previous treatment for prostate cancer with platinum-containing compounds

11. Study Identifiers & Governance

Primary ID:	19839414555992495
Registry Number:	NCT02861573
Sponsor:	Merck
Study Title:	Study of Pembrolizumab (MK-3475) Combination Therapies in Metastatic Castration-Resistant Prostate Cancer (MK-3475-365/KEYNOTE-365)

15. Sign-off & Approval

Principal Investigator: ________________________ Date: _________

Clinical Operations Lead: ______________________ Date: _________

Lead Statistician: ______________________________ Date: _________